

BILL OF MATERIALS

SECTION MATERIALS									
NO	QUANTITY	DESCRIPTION	SIZE	QTY	UNIT CLASS	SHORT CODE	ITEM		
1	PIPE SUELS ASTM A167H101L SCH40S PE	3/4"	1.34	3	538	P	PP0A-00000594		
2	TEE SW CL360S ASTM A167H101L SCH40S PE	3/4"	1	3	538	T	PP0A-00000722		
3	90 DEG ELBOW SW CL360S ASTM A167H101L	3/4"	1	3	538	CPT	PP0A-00000351		
4	90 DEG ELBOW SW CL360S ASTM A167H101L	3/4"	2	538	E		PP0A-00000341		
5	TEE SW 1/2-1/2-1/2-1/2 STRAINER - 1" TYPE	3/4"	1	1			SPECIALTY		
6	TAG NO. 1-1/2-1/2-1/2 STAMPER - 1" TYPE	3/4"	1	338	F		SPECIALTY		
7	SOCKET WELDING FLANGE BR 1/2-1/2-1/2-1/2 SCH40S	3/4"	1	338	G		PP0A-00000468		
8	SPIRAL WOUND GASSET ORING SS 304 CL 150 HOOP SS 1/2 FILLER GRADIENT 3/4"	3/4"	4	538	G		PP0A-00000564		
9	4 CHANNEL UTI HEAVY	1/2"	16	538	B		PP0A-00007682		
10	73 STD BOLL W/CEADUT 1/4" HEAVY HEX NUT ASTM A193B CL2 ASTM A194 UNFUT 1/2"	3/4"	3	538	1A		PP0A-00007760		
11	BALL VALVE SW AN316S CL800 ASTM A318-CP18M TRIM RTFE SEAT LO 3/4"	3/4"	1	538	VC		PP0A-00009607		
12	CHECK VALVE SW CL360S ASTM A167H101L SS316 TRIM STILITE SEATS BC 3/4"	3/4"	1	538	VLST1		PP0A-00008192		
13	BALL VALVE SW W/MPLE SW AN316S CL800 ASTM A318-CP18M TRIM RTFE SEAT FLOATING BALL 3/4" INSECTURE 1/2-1/2-1/2-1/2 SCH40S 3MM LITE	3/4"	4				UBT-G		
14	RTFE (NUT) (M-1016)	3/4"	4				T2-B		
15	12-5/16-1/2-1/2-1/2-1/2	3/4"	2						
CUM PIPE LENGTH		INPD	INPD	INPD	INPD	REMARKS			
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F3047



Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-09-19

LOTTE BIOLOGICS

 LOTTE E&C

PART NO.	AG	PLANT	N	AREA	POB	DWA. NO.	P2AN
M/M NO.	KI-DLSUT-PID-7821-01			PROJ. NO.	16460		HCS
NO.	-			LINE NO.	0.75-HCS-7621602-S98-LH		
NO.						SHT NO.	REV
						1 OF 1	1

VISION	J.Y.KIM	M.H.SUNG	T.H.LIM
ORATION	J.Y.KIM	M.H.SUNG	T.H.LIM
	PREPARED	CHECKED	APPROVED

ISSUED FOR REPAIR	ISSUED FOR INFORMATION	DATE	DESCRIPTION
1	19-09-25		
0	29-08-25		

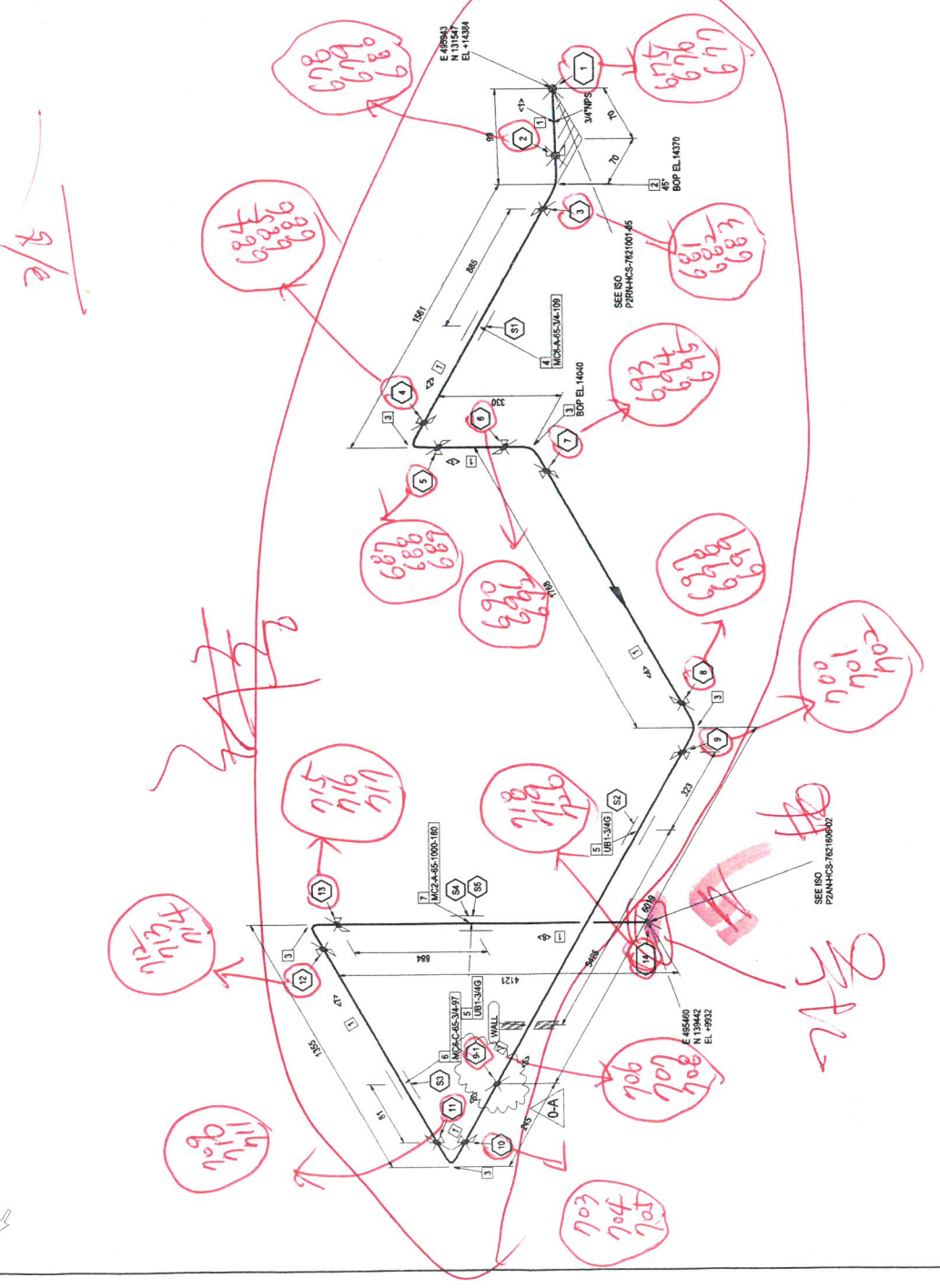
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BILL OF MATERIALS

SECTION MATERIALS						
NO	PRODUCT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE SMLS ACTIVATOR/SHUT OFF VALVES FOR 2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	P	PPE-000000066
2	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	EA	PPE-000000303
3	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
4	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
5	2" OREG. CLEVER SHUT OFF VALVE	3"	2	598	E	PPE-000000344
6	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
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52	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
53	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
54	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
55	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
56	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
57	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
58	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
59	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
60	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
61	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
62	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
63	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
64	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
65	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
66	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
67	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
68	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
69	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
70	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
71	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
72	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
73	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
74	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
75	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
76	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
77	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
78	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
79	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
80	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
81	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
82	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
83	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
84	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
85	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
86	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
87	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
88	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
89	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
90	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
91	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
92	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
93	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
94	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
95	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
96	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
97	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
98	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
99	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344
100	2" OREG. CLEVER SHUT OFF VALVE	3"	1	598	E	PPE-000000344



F2957



Project PROVIDENCE
Approval For Construction
PIPING MANAGER, Bea Chang Won
2025-11-19

PROJECT		PROJECT PROVIDENCE		LOTTE BIOLOGICS	
				LOTTE E&C	
PART NO.	AG	PLANT	N	AREA	POB
PACKED NO.	K1-D5L-JT-PD-782-01			PROD. NO.	16460
DAD NO.				FLUID	HCS
LINE NO.				LINE NO.	0.7HCS-7821606-S98-HH
SWT NO.				SWT NO.	1
REV				REV	0A

[illegible]

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BILL OF MATERIALS

DIRECTION MATERIALS

NO	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PPE SAILS ASTM A133-18/19/20, SCH40S PE	3/4"	21M	S98	P	P82-00000566
2	TEE SW CL2000 ASTM A182-F316/316L	3/4"	2	S98	T	P89-00000572
3	90 DEG ELBOW SW CL2000 ASTM A182-F316/316L	3/4"	3	S98	E	P89-00000541
4	7/8" NO. 1 STD. 1/2" O.D. STAINLESS - F TYPE	3/4"	1	S98	F	SPECIALTY
5	3/4" NO. 1 STD. 1/2" O.D. STAINLESS - F TYPE	3/4"	1	S98	F	SPECIALTY
6	SOCKET WELDING ELBOW SW CL2000 ASTM A182-F316/316L	3/4"	1	S98	E	P89-00000540
7	SOCKET WELDING ELBOW SW CL2000 ASTM A182-F316/316L	3/4"	1	S98	E	P89-00000540
8	70 STD. BOLT W/ NUT HEAVY HEX. NUT ASTM A193-B8 CL2 ASTM A194-B8 UN/UNF	1/2"	16	S98	B	P89-00000502
9	8x 1/4" MM					
10	BALL VALVE SW ANSI/ASTM CL2000 ASTM A182-F316/316L TRIM RTT SEAT FLO	3/4"	3	S98	VL	PP04-00000776
11	ATMOSPHERIC BALL VALVE SW CL2000 ASTM A182-F316/316L TRIM STELLITE SEAT BC	3/4"	1	S98	VC	PP04-00000597
12	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-A
13	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-C
14	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
15	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
16	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
17	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
18	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
19	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
20	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
21	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
22	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
23	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
24	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
25	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
26	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
27	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
28	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
29	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
30	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
31	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
32	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
33	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
34	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
35	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
36	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
37	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
38	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
39	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
40	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
41	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
42	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
43	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
44	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
45	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
46	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
47	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
48	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
49	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
50	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
51	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
52	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
53	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
54	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
55	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
56	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
57	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
58	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
59	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
60	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
61	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
62	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
63	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
64	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
65	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
66	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
67	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
68	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
69	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
70	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
71	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
72	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
73	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
74	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
75	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
76	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
77	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
78	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
79	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
80	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
81	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
82	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
83	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
84	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
85	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
86	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
87	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
88	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
89	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
90	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
91	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
92	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
93	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
94	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
95	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
96	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
97	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
98	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
99	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G
100	PISTON O-RING (NACHTS)	3/4"	2	S98	O	UB1-G

CUT PIPE LENGTH

REMARKS

NO

LENGTH

NO

REMARKS

NO

LENGTH

NO

REMARKS

NO

LENGTH

NO

REMARKS

NO

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NO

REMARKS

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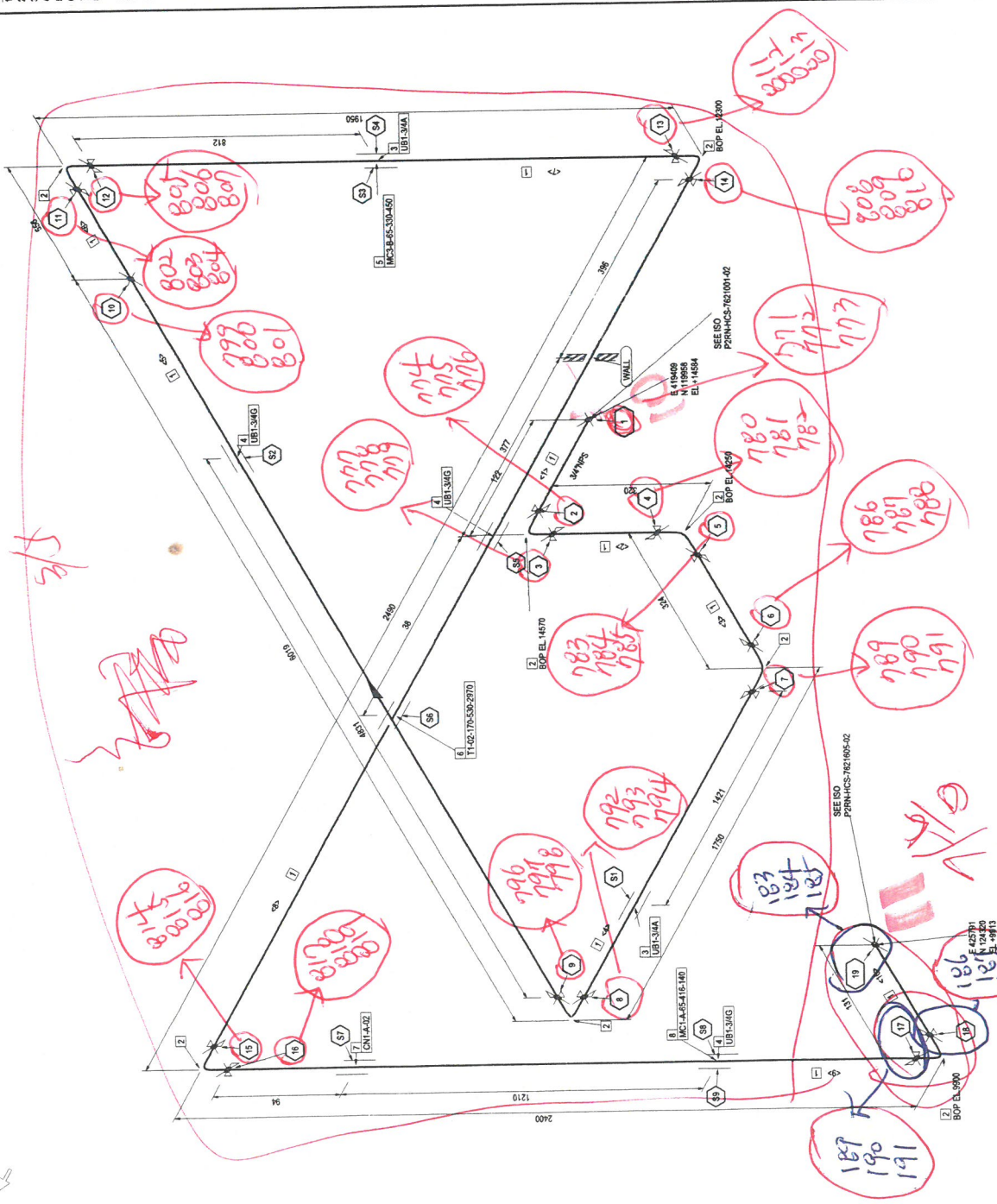
Project PROVIDENCE
Approval For Construction
PIPING MANAGER: Bea Chang Won
2025-08-29

LOTTE BIOLOGICS		LOTTE E&C	
AG	PLANT	AREA	UNIT
K-0801-01-01	N	075-HCS-7621001-01	075-HCS-7621001-01
REV	DATE	REV	DATE
1	2025-08-29	1	2025-08-29

NO	DESCRIPTION	DATE	REV	TEST	INSULATION	PAINT	NOE	STRESS
1	ISSUED FOR INFORMATION	2025-08-29	0					
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BILL OF MATERIALS

SECTION MATERIALS		NO	QUANTITY	UNIT	DESCRIPTION	SIZE	QTY	MAT. CLASS	SHORT CODE	ITEM
		1	1	PIPE	SMLS ASTM A182 F304L SCH40S 95	3"	15.8 M	598	P	PAU-00000566
		2	1	90 DEG	ELBOW (UNION) 2.0000 ASTM A182 F304L	3"	8	598	E	PR9-00002641
		3	2	FLANGE	ASTM A182 F304L	3"	3			UR1-A
		4	1	FLANGE	ASTM A182 F304L	3"	2			UR1-G
		5	1	FLANGE	ASTM A182 F304L	3"	1			MC3-B-V
		6	1	FLANGE	ASTM A182 F304L	3"	1			T1-B
		7	1	FLANGE	ASTM A182 F304L	3"	1			CH1-A-1
		8	1	FLANGE	ASTM A182 F304L	3"	1			MC1-A-V



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PIPE modified at 17:38 31 Jul 2025, Drawing Created at 19:13:55 31 July 2025
PIPE NAME: 0.75-HCS-7621605-S98-1H1_P2RN-01

Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-08-29

LOTTE BIOLOGICS

 LOTTE E&C

PROJECT PROVIDENCE

PART NO.	AG	PLANT	N	AREA	PDB	CWA NO.	PGRN
P&ID NO.	K1-DLSL-JD-7621-01			16480	FLUID		HCS
CAD NO.				LINE NO.	0.75-HCS-7621605-S98-HI		
CWG NO.	P2RN-HCS-7621605-01						1 OF 1
							REV

P2RN-HCS-7621605-01

SH'T NO.	1 OF 1	REV.	0
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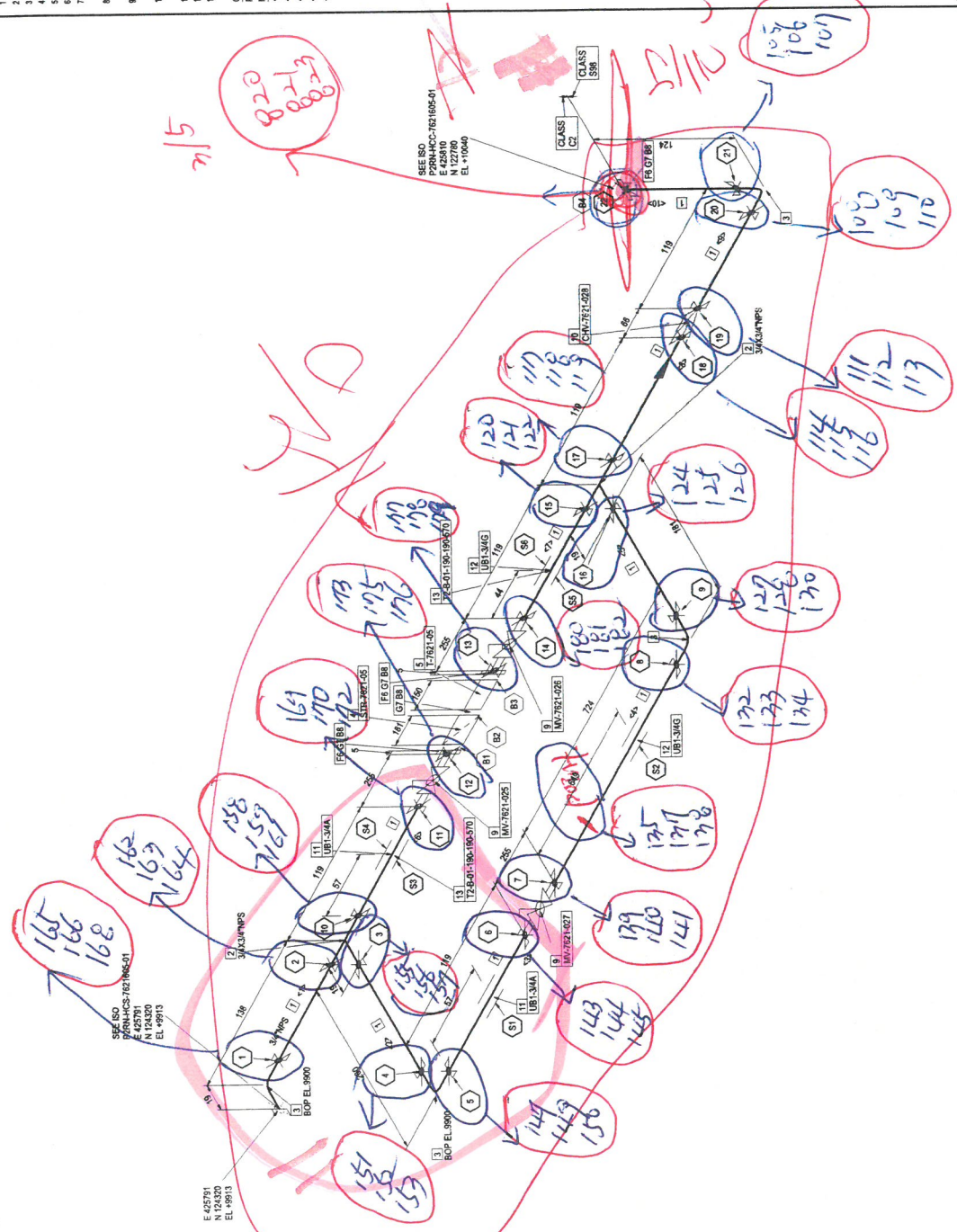
22

F2353



Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-08-29

PROJECT		PROJECT PROVIDENCE		LOTTE BIOLOGICS	
PART NO.		AG	PLANT	N	
P&ID NO.		K-12SL-JUT-PRD-7821-01			
C&D NO.		-			
DRAWING NO.		P&ID NO.		LOTTE E&C	
		P&ID NO.		0.74-HCS-7021605-594-HI	
		FLUID		HCS	
		CWA NO.		P&ID	
		LINK NO.		PRIN	
		SHT NO.		1 OF 1	REV 0

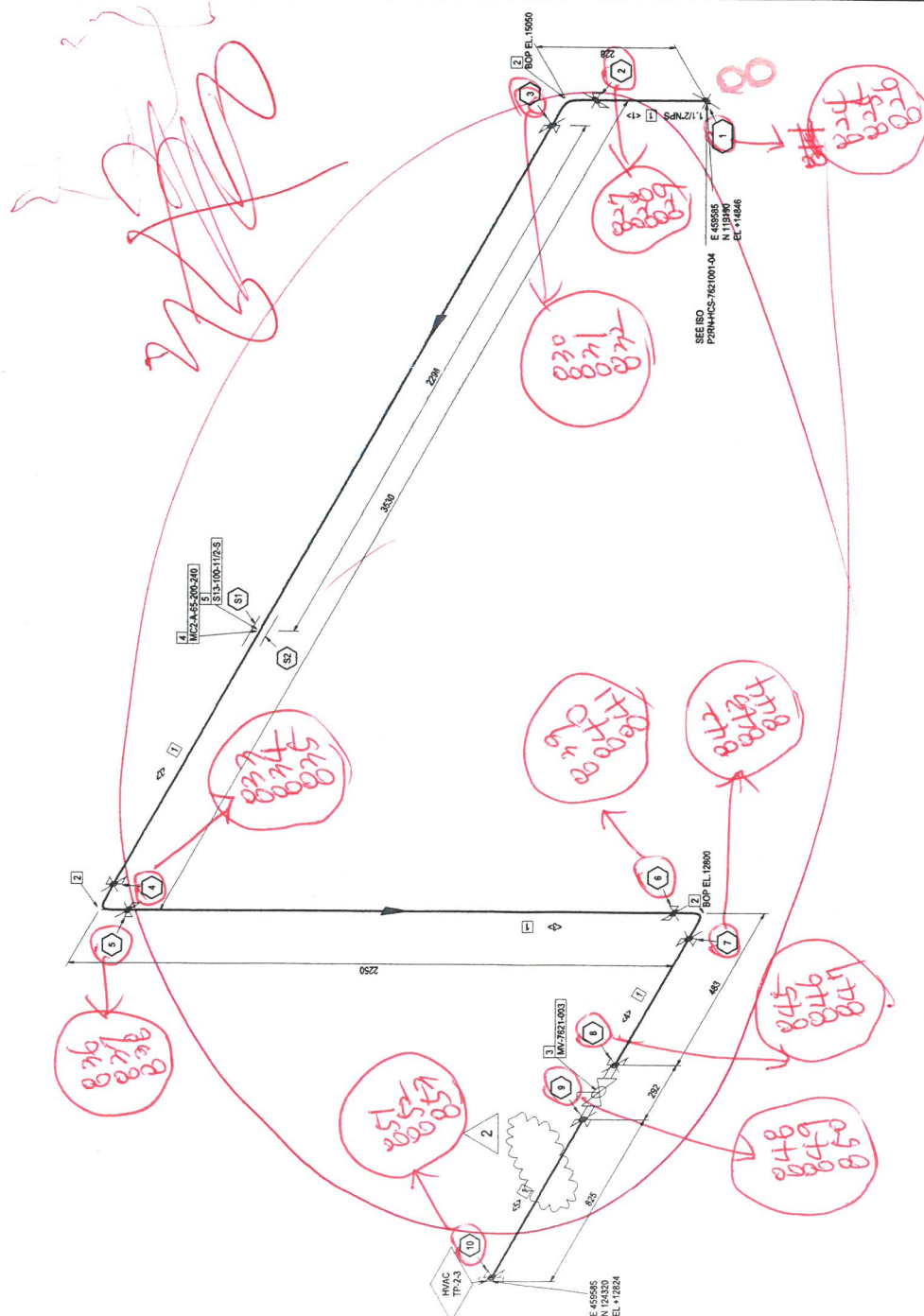
[illegible]

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ERECTION MATERIALS

PIPE COMPONENT DESCRIPTION	SIZE	QTY	MAT. CLASS	SHORT CODE	PLN
1 PIPE S&B ASTM A131X102108L SCHWIS PE	1.02"	7.2 M	538	P	440-00000569
2 90 DEG. ELBOW (IN SCH CLWNO) ASTM A131X108L	1.02"	1	538	E	440-00000569
3 BALL VALVE (IN SCHWIS) ASTM A131X108L CF7M SCH10	1.02"	1	538	VAL	440-00000569
4 ATING BALL FB 3 PCECE BODY (V. OF. (V406967)	1.02"	1			440-00000792
5 MC2-AH-200-240	1.02"				MC2-AH
6 313-001-1053	1.02"				513
CUT PIPE LENGTH					

PRECISE	LENGTH (mm)	NPD (n)	REMARKS	PRECISE	LENGTH (mm)	NPD (n)	REMARKS
<1	196	1.12		<2	367	1.12	
<3	2187	1.12		<4	452	1.12	
<5	818	1.12					

$$\frac{5}{3}$$


F141



Project PROVIDENCE
Approval For Construction
PIPPING MANAGER Bea Chang Won
2025-12-27

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[illegible]

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BILL OF MATERIALS

FABRICATION MATERIALS			
NO	COMPONENT DESCRIPTION	SIZE	QTY
1	PPE GW ASIA MATS 1057H SCHIRS BE RT+100%	2.107	23 M
2	20 GRC GLOW (IN GW NEDIN ASDMPT507H) SCHIRS	2.107	3
FABRICATION MATERIALS			
NO	COMPONENT DESCRIPTION	SIZE	QTY
3	PPE GW ASIA MATS 1057H SCHIRS BE RT+100%	2.107	53 M
4	20 GRC GLOW (IN GW NEDIN ASDMPT507H) SCHIRS	2.107	2
5	G0401	2.107	1
CUT PIPE LENGTH			
NO	COMPONENT DESCRIPTION	SIZE	QTY
6	PPE GW ASIA MATS 1057H SCHIRS BE RT+100%	2.107	53 M
7	20 GRC GLOW (IN GW NEDIN ASDMPT507H) SCHIRS	2.107	2
8	G0401	2.107	1

3

12/2/22

2

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409

Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chiang Won
2025-06-17

PROJECT	LOTTE BIOLOGICS
PROJECT PROVIDENCE	

NO.	AG	PLANT	N	AREA	PDB	CWA NO.	PZRN
NO.	K1-OSL-JUD-7021-01			PROJ. NO.	16480		HCS
				FLUID			
				LINE NO.	2.5-HCS-7621302-S98-PII		

NO.	P2RNHCS-7821302-01	SHT NO.	1 OF 1	REV	0
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DATE	REV.	DESCRIPTION	PREPARED	CHECKED	APPROVED
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THESE RESULTS ARE IN LINE WITH THE CONCLUSIONS OF OTHER STUDIES THAT THE USE OF A SINGLE-STEP METHOD CAN BE AFFECTED BY THE PRESENCE OF A SECONDARY AMINE.

BILL OF MATERIALS

FABRICATION MATERIALS			
NO	COMPONENT DESCRIPTION	SIZE	QTY
1	PIPE END CAP 1/2" SCH 40S 304L	3/4"	1
2	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
3	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
4	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
5	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
6	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
7	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
8	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
9	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
10	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
11	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
12	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
13	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
14	WELDED ELBOW 1/2" SCH 40S 304L	20"	1

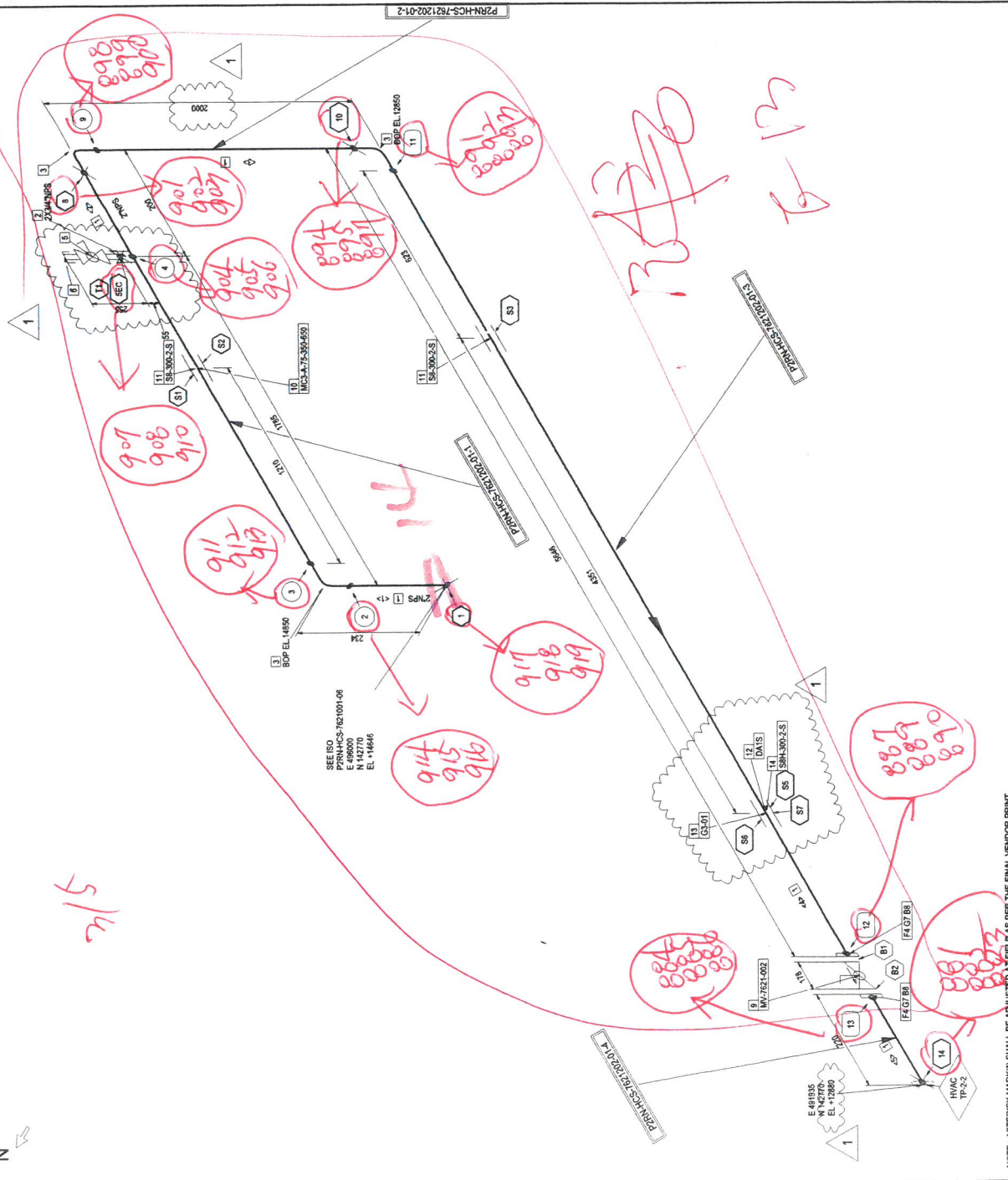
ERECTOR MATERIALS			
NO	COMPONENT DESCRIPTION	SIZE	QTY
1	PIPE END CAP 1/2" SCH 40S 304L	3/4"	1
2	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
3	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
4	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
5	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
6	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
7	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
8	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
9	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
10	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
11	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
12	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
13	WELDED ELBOW 1/2" SCH 40S 304L	20"	1
14	WELDED ELBOW 1/2" SCH 40S 304L	20"	1

408



Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bga Ching Won
2025-11-11

LOTTE BIOLOGICS			
PROJECT PROVIDENCE			
AREA	FLAT	N	
PROJ. NO.	KT-02-01-01		
LINE NO.	2-HCS-7621202-01		
DWG. NO.	P2N-HCS-7621202-01		
SFT. NO.	1 OF 1		
REV.	1-A		



REVISION			
NO.	DESCRIPTION	DATE	BY
1	ISSUED FOR REVISION	11-11-25	
2	ISSUED FOR INFORMATION	11-06-25	
3			
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FABRICATION MATERIALS

ERECTION MATERIALS(NPT)POE (ML1-D16)CUT PIPE LENGTH

PIPING MAN

PROJECT PI

P2RN-H

706



Project PROVIDENCE
Approval For Construction
SIPPING MANAGER Bea Chang Won
2025-12-27



LOTTE BIOLOGICS

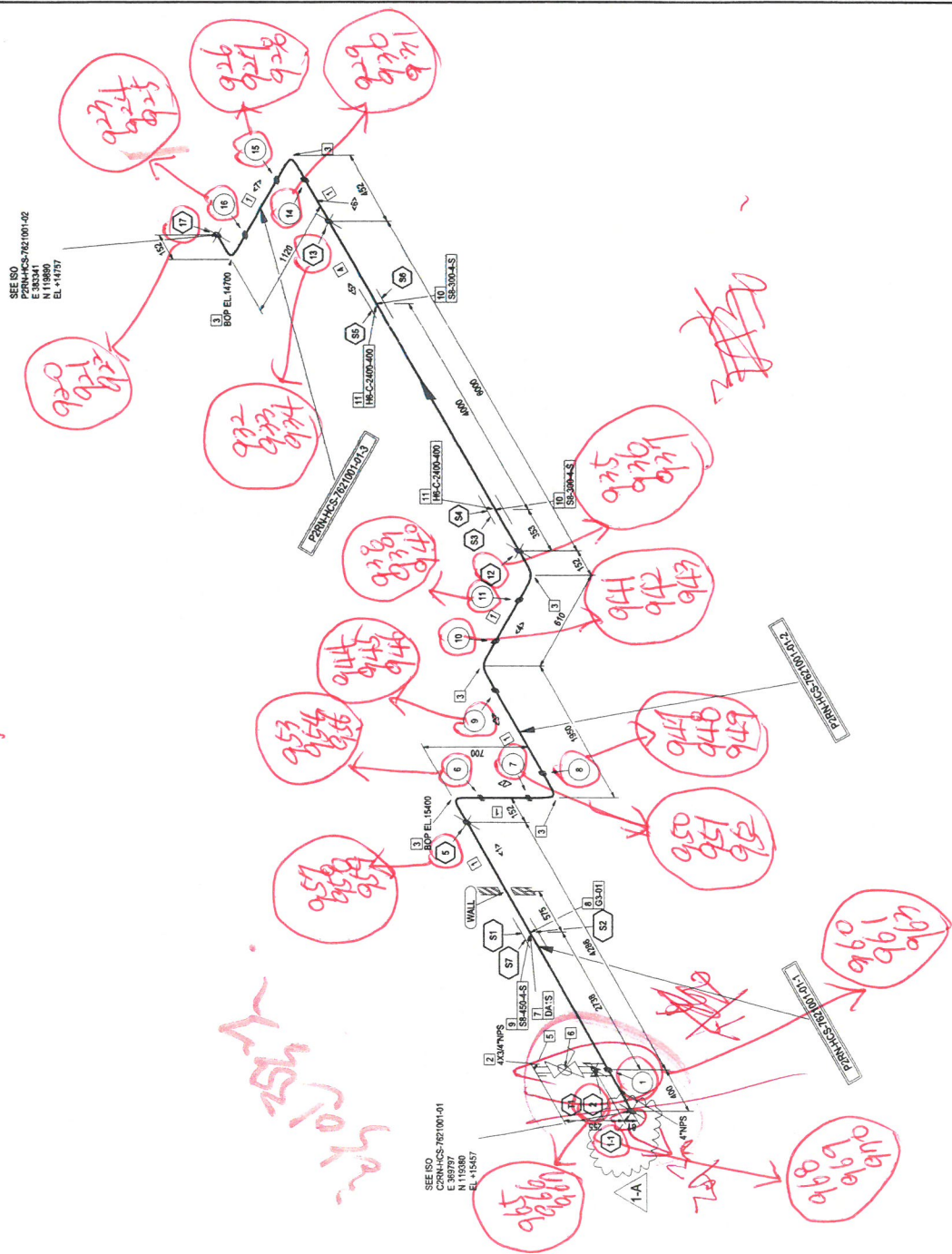
 LOTTE E&C

PROJECT PROVIDENCE

T	AG	PLANT
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D NO	K1-DSL-JT-PID-7621-01
NO	-

NO.	P2RN-HCS-7621001-01
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$$\frac{5}{2}$$


NOTE : ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

[illegible]

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BILL OF MATERIALS

FABRICATION MATERIALS				
NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT. CLASS
1	PPE ASTH A152-11631HL, SCH10S RE-100%	4"	538	P
2	PPE ASTH A152-11631HL, SCH10S RE-100%	4"	0.2 M	P
3	HALF COUPLER SW CL2000 ASTH A152-11631HL	3/4"	1	538
4	REDUCING TEE SW WELDED ASTH A152-11631HL, SCH10S	4X2"	1	538
5	90 DEG. ELBOW (LW) WELDED ASTH A152-11631HL, SCH10S	3"	1	538
6	HALF COUPLER SW CL2000 ASTH A152-11631HL, SCH10S	3/4"	1	538
7	CAP SW 1/4" SAE 1501 ASTH A152-11631HL, SCH10S	3"	1	538
SECTION MATERIALS				
NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT. CLASS
1	PPE ASTH A152-11631HL, SCH10S RE-100%	4"	35.0 M	P
2	PPE SW 1/4" SAE 1501 ASTH A152-11631HL, SCH10S	3/4"	0.1 M	P
3	CAP 5/8" NPT CL2000 ASTH A152-11631HL	3/4"	1	538
4	90 DEG. ELBOW (LW) SW CL2000 ASTH A152-11631HL	3"	1	538
5	BALL VALVE SW WAMPPE ANS150.1 CL2000 ASTH-CPRM SS316 TRIM RITE	3/4"	1	538
6	TEST FLOATING BALL 3/4" PRICE BODY 1/4" OP. A152-11631HL, SCH10S TRIM RITE	3/4"	1	538
7	POPE (L-101F)			VLS1
8	CO-41	4"	2	G3
9	SA-3000-S	4"	3	38
10	DAIS	6"	1	38
11	SA-464-S	6"	1	38
FABRICATION MATERIALS				
NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT. CLASS
1	PPE ASTH A152-11631HL, SCH10S RE-100%	4"	538	P
2	PPE ASTH A152-11631HL, SCH10S RE-100%	4"	0.2 M	P
3	HALF COUPLER SW CL2000 ASTH A152-11631HL	3/4"	1	538
4	REDUCING TEE SW WELDED ASTH A152-11631HL, SCH10S	4X2"	1	538
5	90 DEG. ELBOW (LW) WELDED ASTH A152-11631HL, SCH10S	3"	1	538
6	HALF COUPLER SW CL2000 ASTH A152-11631HL, SCH10S	3/4"	1	538
7	CAP SW 1/4" SAE 1501 ASTH A152-11631HL, SCH10S	3"	1	538
SECTION MATERIALS				
NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT. CLASS
1	PPE ASTH A152-11631HL, SCH10S RE-100%	4"	35.0 M	P
2	PPE SW 1/4" SAE 1501 ASTH A152-11631HL, SCH10S	3/4"	0.1 M	P
3	CAP 5/8" NPT CL2000 ASTH A152-11631HL	3/4"	1	538
4	90 DEG. ELBOW (LW) SW CL2000 ASTH A152-11631HL	3"	1	538
5	BALL VALVE SW WAMPPE ANS150.1 CL2000 ASTH-CPRM SS316 TRIM RITE	3/4"	1	538
6	TEST FLOATING BALL 3/4" PRICE BODY 1/4" OP. A152-11631HL, SCH10S TRIM RITE	3/4"	1	538
7	POPE (L-101F)			VLS1
8	CO-41	4"	2	G3
9	SA-3000-S	4"	3	38
10	DAIS	6"	1	38
11	SA-464-S	6"	1	38

PIECE	LENGTH	NPD	REMARKS	PIECE	LENGTH	NPD	REMARKS
0	(MM)	(IN)		NO	(MM)	(IN)	

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6000	4	6000	4
8000	4	6000	4

	6000	4	<6>	5984	4
33	6000	4	<6>	5984	4
17	150	3	<6>	100	3M
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Project PROVIDENCE

Approval For Construction
PIPING MANAGER Bea Chang Won

2025-11-25

LOTTE BIOLOGICS

PROJECT PROVIDENCE

[illegible]

AG	PLANT	N	AREA	PDB	CWA NO.	P2RN
K1-DSL-JT-PIID-7921-01						
					16480	HCS

LINE NO.	4-HCS-7621001-S98-IHI	SH'T NO	1 OF 4
0.	P2RN-HCS-7621001-02		

1-A	REV	1 OF 1	SHEET NO.
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[illegible]

THE EFFECT OF THE INTENSITY OF IRRADIATION OF LEC.

NOTE : ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT

Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-11-25



LOTTE BIOLOGICS



LOTTE E&C

CWA NO.	P2RN
FLUID	HCS
1001-S98-IH	
SHT NO.	1 OF 1
REV	
1-A	

908



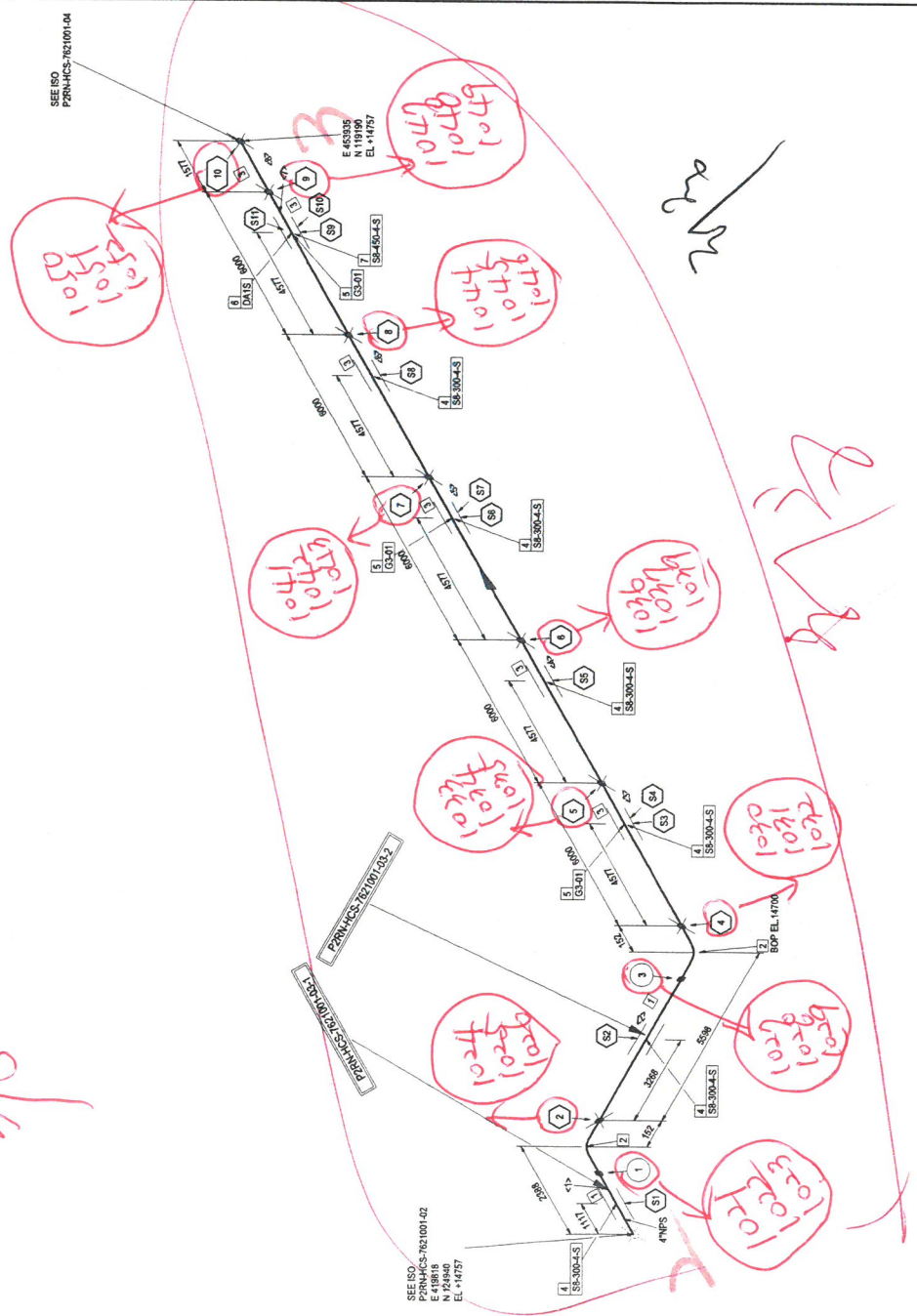
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PIPE NAME: 4HCS-7621001-S98-H1_P2RN-02

609



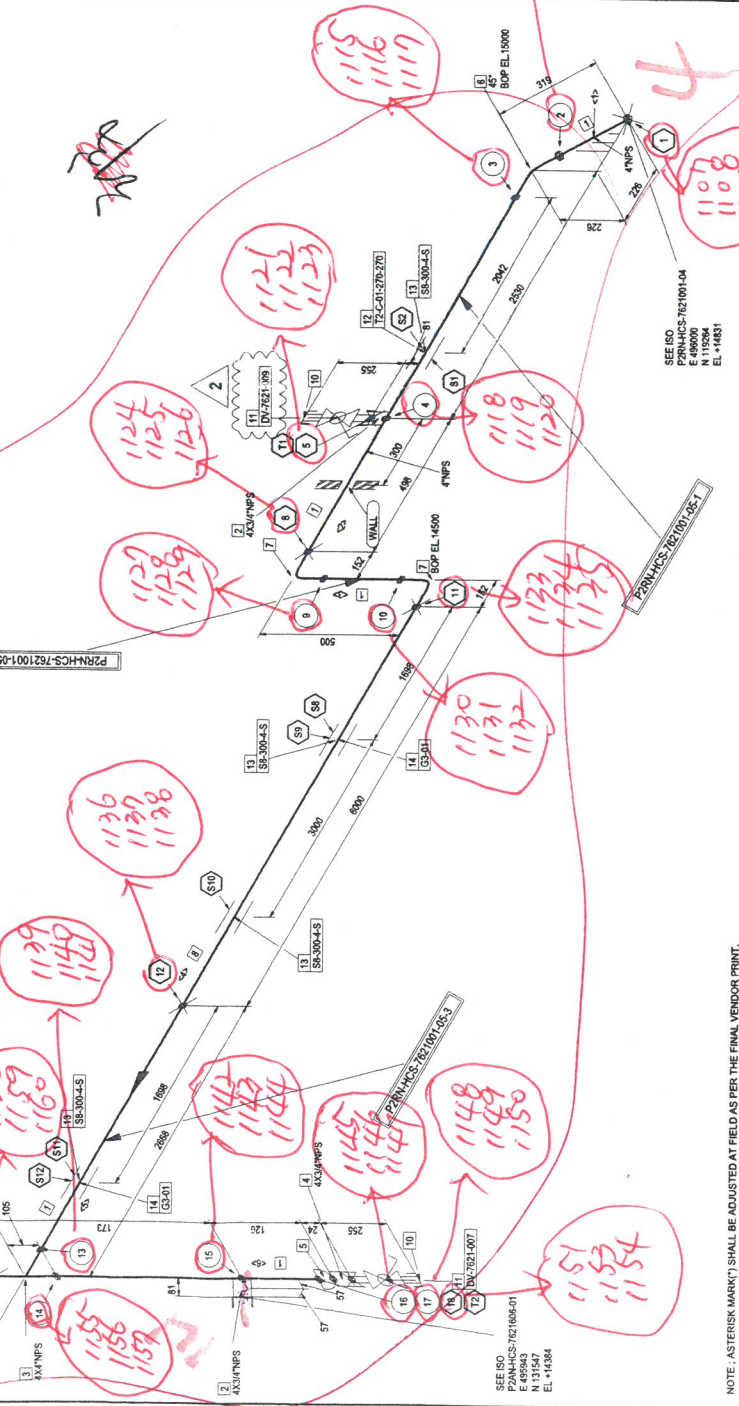
Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-07-15

PROJECT		PROJECT PROVIDENCE		LOTTE BIOLOGICS	
ART		AG	PLANT	CW	PDB
K1-DSU-JUT-PD-762-01					
ZAO		-			
LINE NO.					
P2R2H-HCS-762-1001-03					
SHT NO.					
1 OF 1					
REV					
0					

[illegible]

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BILL OF MATERIALS

[illegible]

Project PROVIDENCE
Approval For Construction
SIPING MANAGER Bea Chang Won
2026-01-17



119

PROJECT		PROJECT PROVIDENCE		LOTTE BIOLOGICS	
K108-JLF-PID-3241-61		N		LOTTE E&C	
PI	AG	SLANT	AREA	PDB	CWA NO
NO	NO	NO	NO	16480	FLUID
NO	NO	NO	LINE NO	44PCS-67101-3834	PRRN HCS
NO	P2RN-HCS-782101-05			SHT NO	2

[illegible]

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SEE ISO
P2AN-HCS-7
E 495943
N 131547
EL +14384

SEE ISO
228N-HCS-7621001-04

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[illegible]

SECTION MATERIALS	NO	COMPONENT DESCRIPTION	SIZE	QTY	MAT. CLASS	SHORT CODE	ITEM
	1	PIPE ASTM A312 304/316L SCH 80S BE RT 100%	4"	13.1 M	538	P	PAC-00001364
	2	PIPE FLANGE ASTM A182 F316L	3"			CPT	PAC-00003801
	3	SPRING WOUND GASKET CARG. 50 CL 1/2 HOOP	2	1	538	FB	PRI-00000419
	4	E LAMINATED GASKET	COUPLER 2"	2	538	G	PRI-00000657
	5	90 STD BOLT 1/2" X 4" HEAVY HEX M16 X 40	50"	0	538	B	PRI-00001892
	6	BALL VALVE FLG 1/2" ASTM A194 B8C 150 LB	2"	1	538	VA	PRI-00001892

[illegible]

	(M)	(N)	NO	(M)	(N)
1	2211	4	<2	446	4
2	6000	4	<6	1053	4
3	6000	4	<6	6000	4
4	6000	4	<6	2019	4
5	88	2	<10	452	4
6	272	3			

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Project PROVIDENCE


 Project Approved By
PIPING MANAGER, Bae Chang Won
2025-09-17
 Approval For Construction
 Date
 09 Aug 2025 Drawing Created at 11:01 AM

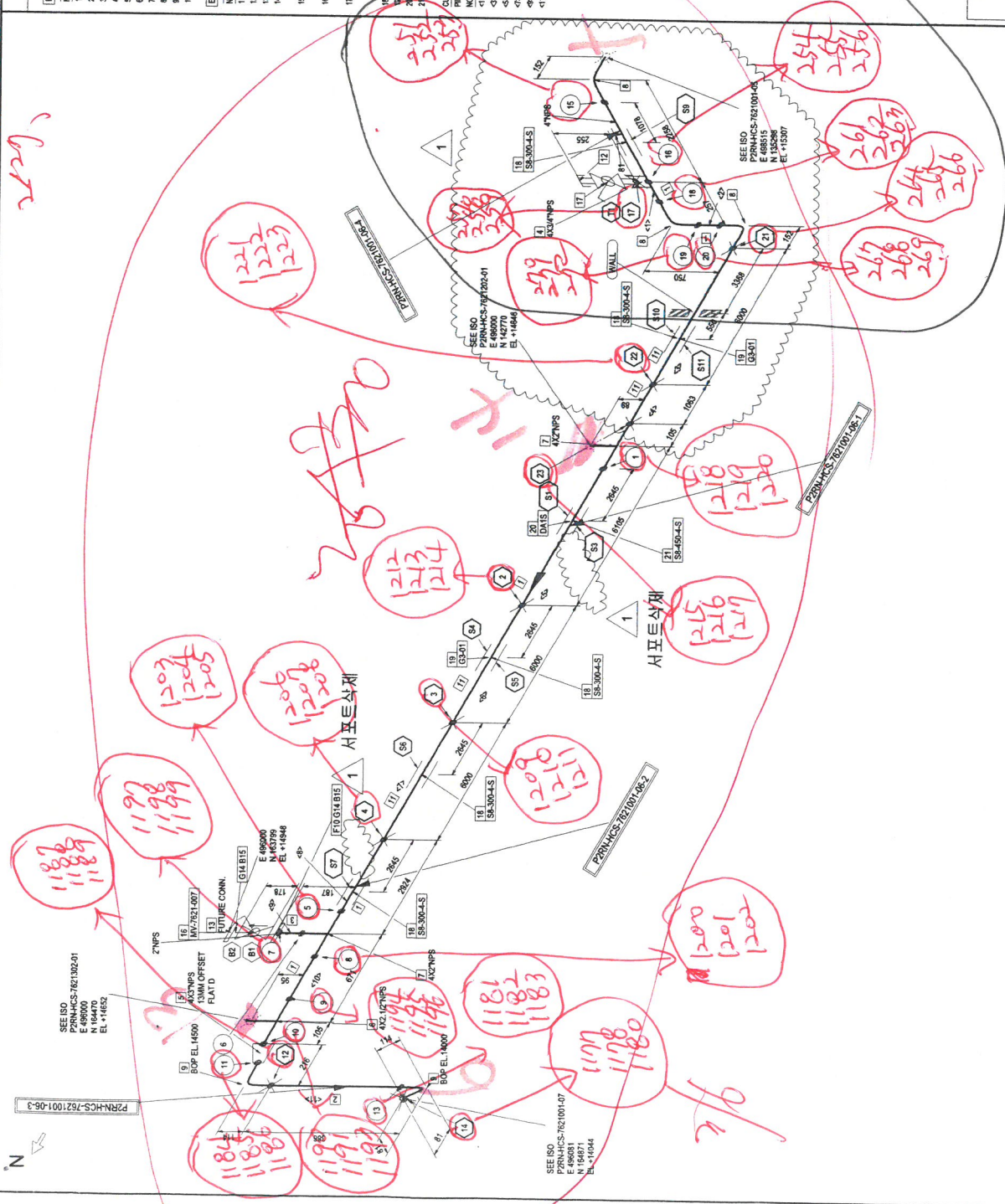
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AG	K1-DLSL-JT-PID-7621-01		16400	FLUDD	HCS
LINE NO.			4-HCS-7621001-596-H		
1 OF 1					
REV					

Project PROVIDENCE
Approval For Construction
SHIPING MANAGER, Bea Chang Won
2025-09-17


LOTTE BIOLOGICS LOTTE E&C

P2RN-HCS-7621001-06

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PROJECT

100

REV	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
2A	29-09-25				
2	17-06-25	ISSUED FOR REVISION	JYKM	MAJSLG	TALIM
			MAJSLG	TALIM	
1		ISSUED FOR REVISION	JYKM	MAJSLG	TALIM
			MAJSLG	TALIM	
0	15-07-25	ISSUED FOR INFORMATION	JYKM	MAJSLG	TALIM
			MAJSLG	TALIM	

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BILL OF MATERIALS

FABRICATION MATERIALS

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PIPE END ASTM A152.11018L SCH10S BE RT-100%	3"	0.3M	S98	P	PA2-00001803
2	HAIF COUPLING SW CL3000 ASTM A182-F15218L	3X3/4"	1	S98	TH	PP9-00003891
3	TEE SW WELDED ASTM A182-F15218L SCH10S	3X3"	1	S98	T	PP9-00003891
4	HAIF COUPLING SW CL3000 ASTM A182-F15218L	3X3/4"	1	S98	TH	PP9-00003891
5	CAP SW SMLS ASTM A182-F15218L SCH10S	3"	1	S98	CP	PP9-00003891
6	SUP-ON FLANGE RF CL150 ASTM A182-F15218L	3"	1	S98	F	PP9-00003891

SECTION MATERIALS

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
7	CAP SORD (NPT) CL3000 ASTM A182-F15218L	3/4"	1	S98	CPT	PP9-00003891
8	BLIND FLANGE RF CL150 ASTM A182-F15218L	3"	1	S98	FB	PP9-00004020
9	SPRAL WOUND GASKET O-RING SS 3M CL150 HOOP SS 316 FILLER GRAPHIT	3"	2	S98	G	PP9-00006058
10	1/2 STD BOLT 1/2" HEAVY HEX NUT ASTM A193-B8 CL2 ASTM A194-B8 UNF	5/8"	8	S98	B	PP9-00019599
11	BALL VALVE FLO-JOIF CL150 ASTM A351-CF8M SS316 TRIM FITE SEAT FLOATING	3"	1	S98	VL	PP9-00007541
12	BALL VALVE SW WAMPLE ANSICR CL3000 ASTM A351-CF8M SS316 TRIM FITE SEAT FLOATING BALL FB 3 PRCE BODY LV OF A152.11018L SCH10S 7MM TDE (NPT) (N-1-D16)	3/4"	1	S98	VLSTI	PP9-00008192

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
13	PIPE END ASTM A152.11018L SCH10S BE RT-100%	3"	0.3M	S98	P	PA2-00001803
14	HAIF COUPLING SW CL3000 ASTM A182-F15218L	3X3/4"	1	S98	TH	PP9-00003891
15	TEE SW WELDED ASTM A182-F15218L SCH10S	3X3"	1	S98	T	PP9-00003891
16	HAIF COUPLING SW CL3000 ASTM A182-F15218L	3X3/4"	1	S98	TH	PP9-00003891
17	CAP SW SMLS ASTM A182-F15218L SCH10S	3"	1	S98	CP	PP9-00003891
18	SUP-ON FLANGE RF CL150 ASTM A182-F15218L	3"	1	S98	F	PP9-00003891

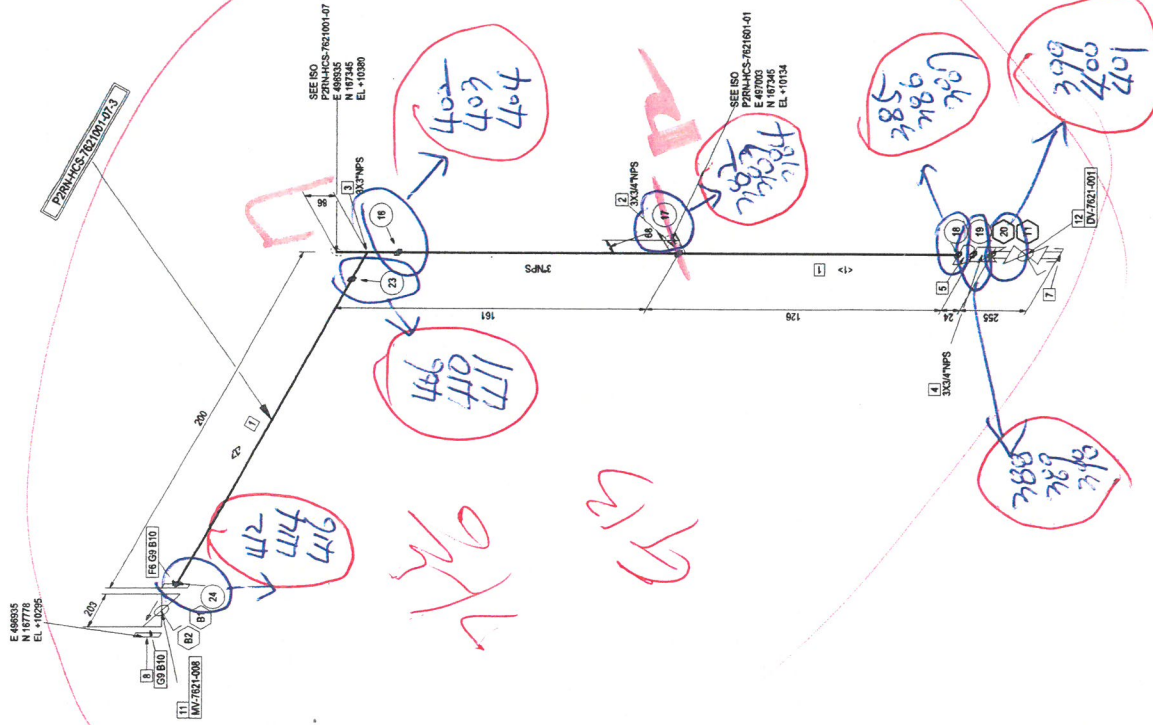
P2RN-HCS-7621001-07에서 분리된 ISO링

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Project PROVIDENCE
Approval For Construction
PIPING MANAGER : Beo Chang Won
2025-09-29

PROJECT		LOTTE BIOLOGICS		LOTTE E&C	
PROJECT PROVIDENCE		LOTTE BIOLOGICS		LOTTE E&C	
PART	NO.	AG	PLANT	N	AREA
P&ID NO.	K1-051-JT-PID-7621-01				
PROJ NO.	16480				
LINE NO.	4-HCS-7621001-598-HI				
DATE	2025-09-29				
REV	1				
REV	1				
REV	1				



NOTE: (ASTERISK MARK) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

MATERIAL CLASS		DESIGN		TEST		STRESS	
NPS (inch)	MATL CLASS	DESIGN PRESS. (bar/g)	DESIGN TEMP. (°C)	TEST PRESS. (bar/g)	TEST TEMP. (°C)	STRESS (MPa)	STRESS (ksi)
3	S98	144	144	172	172	172	172
0.75	S98	144	144	172	172	172	172

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EXTRACTION MATERIALS

COMPONENT DESCRIPTION	SIZE	QTY	UNIT CLASS	SHORT CODE	ITEM
PIPE S&S ACT 14.5 IN DIA SCH 40 R	3"	1.1 M	58		00000584
80-855 ELBOW 1/2 IN SCH 40 ACT 14.5 IN DIA	3"	1	58	E	00000584
MO-2-65-34-450	3"	1			MO-2-65-34-450
MO-2-65-34-720	3"	1			MO-2-65-34-720
MO-2-65-34-792	3"	1			MO-2-65-34-792

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Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2025-06-24

LOTTE BIOLOGICS

PROJECT PROVIDENCE

	AG	PLANT	
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NO.	.
NO.	PSFENJHCS-7621401-01

HI				
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0.75-HCS-7621401-S98-I	SU'T NO
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LINE NO.	
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PHCS-7621401-01

NO.	
NO.	PSEA

APPROVED	THLIM	CAD
REJECTED		DWG

	H.S. JUNG	M.H.
	PREPARED	CHE

DESCRIPTION	UNITED FOR INFORMATION
1. The first part of the report is a description of the project and the objectives of the study.	
2. The second part of the report is a description of the methodology used in the study.	
3. The third part of the report is a description of the results of the study.	
4. The fourth part of the report is a description of the conclusions of the study.	
5. The fifth part of the report is a description of the recommendations of the study.	
6. The sixth part of the report is a description of the limitations of the study.	
7. The seventh part of the report is a description of the future research.	
8. The eighth part of the report is a description of the references.	
9. The ninth part of the report is a description of the appendices.	
10. The tenth part of the report is a description of the index.	

DATE	C3-00-4	1230
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REV	0	2
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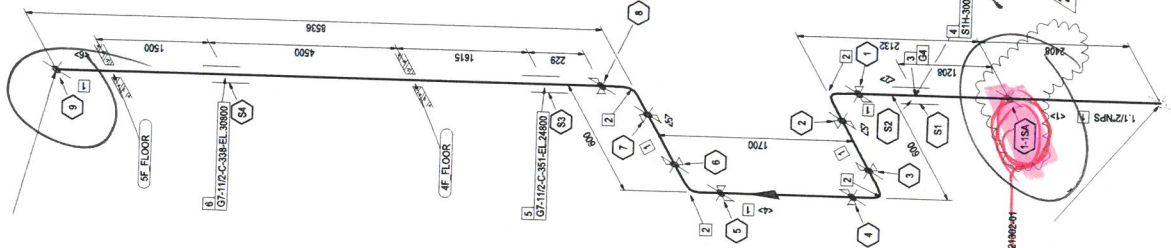
SEE ISO
PSEN-HCS-7621401-02

SEE ISO
227RN-HCS-7621401-01
E 504116
4 113245
1 +15913

NOTE : ASTERISK MARK(*) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

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SEE ISO
PSEN-HCS-7621502-02
E 504235
N 154470
EL 118184



BILL OF MATERIALS

ERECTOR MATERIALS

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	15.4	M	P	P80-00000000
2	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	4	E	E	P80-00000000
3	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	1	E	E	P80-00000000
4	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	1	E	E	P80-00000000
5	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	1	E	E	P80-00000000
6	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	1	E	E	P80-00000000

CUT PIPE LENGTH

NO.	COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	15.4	M	P	P80-00000000
2	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	4	E	E	P80-00000000
3	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	1	E	E	P80-00000000
4	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	1	E	E	P80-00000000
5	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	1	E	E	P80-00000000
6	PIPE ELBOW (ASTM A182 F316L SCH 40)	1.12"	1	E	E	P80-00000000

F1229



Project PROVIDENCE
Approval For Construction
PIPING MANAGER Bea Chang Won
2026-01-30

PROJECT		LOTTE BIOLOGICS	
PROJECT PROVIDENCE		LOTTE E&C	
PART	AG	PLANT	N
PAID NO	K1-00000000-001	AREA	P20
QAD NO	-	PROJ NO	16400
LINE NO	-	LINE NO	15-HCS-7621502-S98-HI
ENG NO	-	SHT NO	1 OF 1
REV	2-A	REV	2-A

NOTE: (ASTERISK MARK) SHALL BE ADJUSTED AT FIELD AS PER THE FINAL VENDOR PRINT.

NPS (Inch)	MATL CLASS	OFFER	DESIGN	TEST	STRESS
1.5	S98	PRESS: 3 TEMP: 144 (bar.g)	PRESS: 4.0 TEMP: 172 (bar.g)	PRESS: 6 TEMP: 172 (bar.g)	1.172/2/PS

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BILL OF MATERIALS

ELECTION MATERIALS		NO		COMPONENT DESCRIPTION	SIZE	QTY	MATL CLASS	SHORT CODE	ITEM
1				PPE SALS ASTM A12-304/16L SCH16S PE	1.107	15.1 M	S36	P	PA02-20000568
2				90 DEG. ELBOW (1/2" SW CL 3000 ASTM A16-F143 RL	1.107	4	S36	E	PA02-20000564
3				90	1.107	1			GA
4				SHH-200-112-304H-13L11	1.107	1	S1-S		S1-S
5				GT-110-C-38-EL-2000	1.107	1	GT-C		GT-C
6				GT-110-C-38-EL-2000	1.107	1	GT-C		GT-C

CUT PIPE LENGTH		NPD		LENGTH		PRICE		REMARKS	
NO	(M)	(M)	(M)	NO	(M)	NO	(M)	NO	(M)
<1>	408	1.102	<2>	537	1.102				
<3>	1837	1.102	44>	537	1.102				
	8565	1.102							

F1229

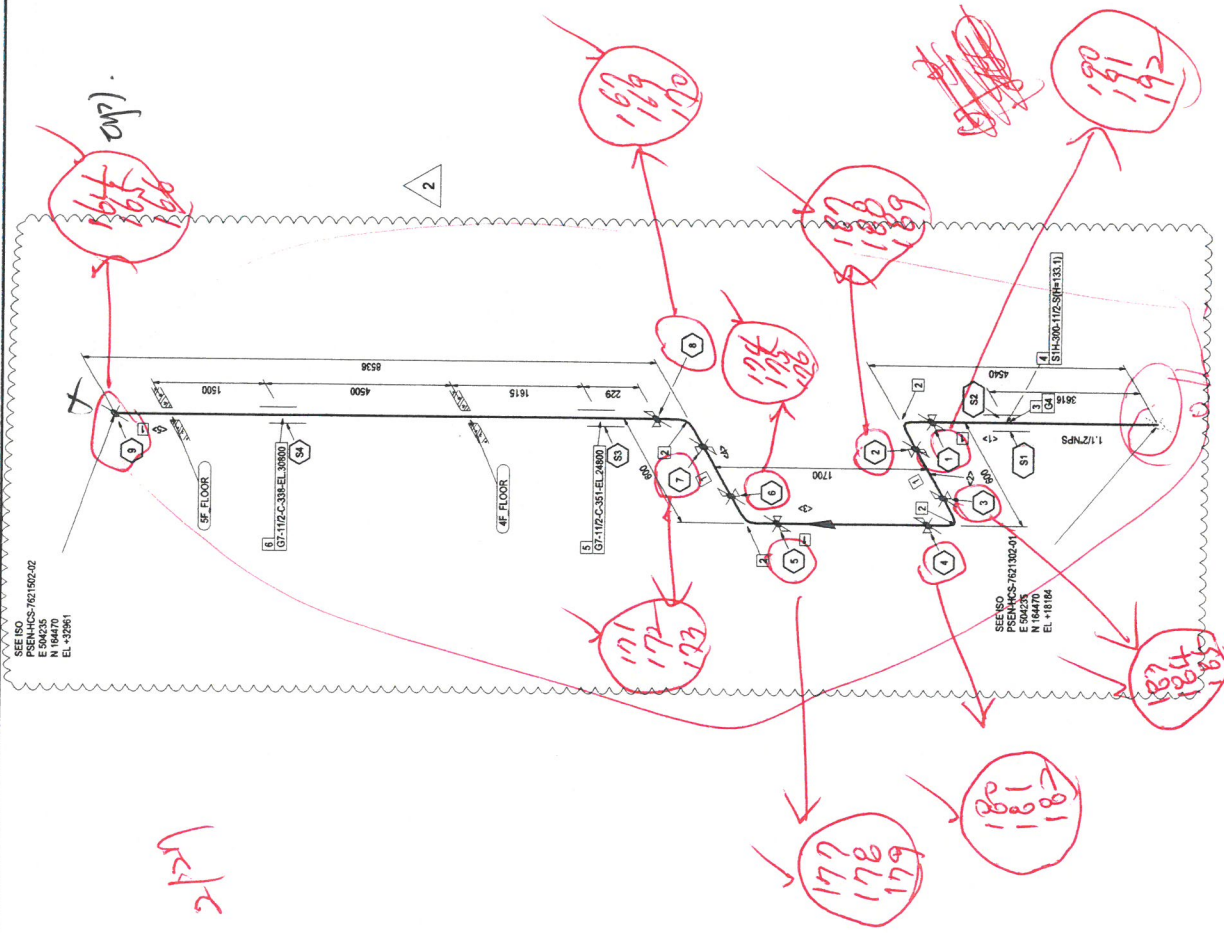


Project PROVIDENCE
Approval For Construction
PIPING MANAGER, Bea Chang Won
2025-09-19

LOTTE BIOLOGICS



PARCEL	AG	PLANT	N	AREA	POB	DWA NO	PSEN
PAID NO	K1-JSL-JT-PD-762-101			FLUID	16480		HCS
CAD NO.	-			LINE NO.	1.5-HCS-7621502-S98-HH		
PSEN-HCS-7621502-01							1 OF 1
DWG NO.						SHT NO.	REV
							2



NPS (inch)	M&T CLASS	OPER.		DESIGN		TEST		INSULATION THK (mm)	PAINT CODE	PWHT (RT/2)	STRESS	REV	DATE	DESCRIPTION
		PRESS. (lb/g)	TEMP. (°F)	PRESS. (bar/g)	TEMP. (°C)	MEDIUM	TYPE							
1.5	S88	3	144	4.0 FV	172	6	H	BH 25	-	N	-	2	19-09-25	ISSUED FOR REVISION
												1		ISSUED FOR REVISION
												0	15-07-25	ISSUED FOR REVISION
														ISSUED FOR INFORMATION

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944



Project PROVIDENCE
Approval For Construction
SUPERVISING MANAGER: Bea Chang Won
2025-09-19

LOTTE BIOLOGICS

 LOTTE & CO.

CWA NO.	PSEN
FLUID	HCS

21302-S98-IH1

SHT NO.	1 OF 1	REV	1
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FABRICATION MATERIALS		NO		COMPONENT DESCRIPTION	SIZE	QTY	MAT. CLASS	SHORT CODE	ITEM
1		1		PIPE WELD ASTM A312-316/317L SCH15 BS RT-100%	2.1/2"	2.1 M	538		PAC6-00001112
2		2		REDUCING TEE BW WELDED ASTM A403-WP316/317L SCH15 SCH15	2.1/2"	1	538	TR	P959-00003148
3		3		CONV. SMOKE BLEPDE SMLS ASTM A403-WP316/317L SCH15 SCH15	2.1/2X1.0/2"	1	538	RCS	P959-01010558
EJECTION MATERIALS		NO		COMPONENT DESCRIPTION	SIZE	QTY	MAT. CLASS	SHORT CODE	ITEM
4		4		90 DEG. ELBOW (BW WELDED) ASTM A403-WP316/317L SCH15	2.1/2"	1	538	E	P959-00001112
5		5		GT-210-A-038-L-1/2IN	2.1/2"	1			GT-A
6		6		STP-212-A-30-05-EW	2.1/2"	1			STP1
CUT PIPE LENGTH		PRICE	LENGTH	REMARKS	NPD	NO	REMARKS	NPD	REMARKS
	PRICE	(RM)	(MM)		(RM)			(RM)	
			2002	2.1/2"					

[illegible]

